

U(1) - Symmetric Fermionic Gaussian Transfer Matrix Formalism

Hamiltonian - to - Covariance Map

Generic U(1) - symmetric Gaussian state: $\hat{\rho} = \frac{1}{Z} e^{-\sum_{ij} h_{ij} \hat{c}_i^\dagger \hat{c}_j}$, $h = h^\dagger \in \mathbb{C}^{N \times N}$ (N modes)

Diagonalize $h = U D U^\dagger$, where $U \in U(N)$.

Define the many-body unitary: $\hat{U}_U$ with single-particle action

$$\hat{U}_U \hat{c}_i^\dagger \hat{U}_U^{-1} = \sum_j \hat{c}_j^\dagger U_{ji}, \quad \hat{U}_U \hat{c}_i \hat{U}_U^{-1} = \sum_j (U^\dagger)_{ij} \hat{c}_j$$

$$\text{Then } \sum_{ij} h_{ij} \hat{c}_i^\dagger \hat{c}_j = U_{ik} D_{kl} (U^\dagger)_{lj} \hat{c}_i^\dagger \hat{c}_j = (\hat{c}_i^\dagger U_{ik}) D_{kl} (U^\dagger)_{lj} \hat{c}_j, \quad D_{kl} \equiv \delta_{kl} \lambda_k$$

Now, define the quasiparticles $\hat{\chi}_i^\dagger = \sum_j \hat{c}_j^\dagger U_{ji}$, $\hat{\chi}_i = \sum_j (U^\dagger)_{ij} \hat{c}_j$

$$\Rightarrow \hat{\rho} = \frac{1}{Z} e^{-\sum_k \lambda_k \hat{\chi}_k^\dagger \hat{\chi}_k}, \quad \text{Tr}(\hat{\rho}) = \frac{1}{Z} \prod_k (1 + e^{-\lambda_k}) = 1, \Rightarrow Z = \prod_k (1 + e^{-\lambda_k}) = \prod_k z_k$$

$$\langle \hat{c}_n^\dagger \hat{c}_m \rangle = \text{Tr}(\hat{\rho} \hat{c}_n^\dagger \hat{c}_m) = \frac{1}{Z} \text{Tr}(e^{-\sum_k \lambda_k \hat{\chi}_k^\dagger \hat{\chi}_k} \hat{c}_n^\dagger \hat{c}_m)$$

$$\text{Use } \hat{c}_i^\dagger = \sum_j \hat{\chi}_j^\dagger (U^\dagger)_{ji}, \quad \hat{c}_i = \sum_j (U)_{ij} \hat{\chi}_j$$

$$\Rightarrow \langle \hat{c}_n^\dagger \hat{c}_m \rangle = \frac{1}{Z} \text{Tr}(e^{-\sum_k \lambda_k \hat{\chi}_k^\dagger \hat{\chi}_k} \sum_{n'} \hat{\chi}_{n'}^\dagger (U^\dagger)_{n'n} \sum_{m'} (U)_{mm'} \hat{\chi}_{m'})$$

$$= \sum_{n'm'} (U^\dagger)_{n'n} (U)_{mm'} \delta_{n'm'} (e^{-\lambda_{n'}} / Z_{n'}), \quad \text{define } \xi_k = \frac{1}{1 + e^{-\lambda_k}}, \quad \Lambda_{kl} = \delta_{kl} \xi_k$$

$$\Rightarrow \langle \hat{c}_n^\dagger \hat{c}_m \rangle = \sum_{n'm'} (U^\dagger)_{n'n} U_{mm'} \Lambda_{n'm'} = (U^* \Lambda U^T)_{nm} \quad \begin{array}{l} \xi_k \in [0, 1], \quad \lambda_k \rightarrow \infty \Rightarrow \xi_k \rightarrow 0 \\ \lambda_k \rightarrow -\infty \Rightarrow \xi_k \rightarrow 1 \end{array}$$

The 2-point function will be denoted as $C_{nm} \equiv \langle \hat{c}_n^\dagger \hat{c}_m \rangle$

$$\text{Complex Covariance: } G = 2C - 1 \sim 2\Lambda - 1 = \frac{2}{1 + e^D} - 1 = \frac{2 - 1 - e^D}{1 + e^D} = \frac{e^{-D/2} - e^{D/2}}{e^{-D/2} + e^{D/2}}$$

Since $h = (U^* D U^T)^T$, it follows that

$$\Rightarrow 2\Lambda - 1 = -\tanh(D/2)$$

$$\Rightarrow \underline{G = -\tanh(h^T/2)}, \quad \text{Where } G_{nm} = \langle [\hat{c}_n^\dagger, \hat{c}_m] \rangle$$

We will use the complex covariance to label the state.

State - Covariance Relation (pure state)

Consider a generic free-fermion pure state with definite global charge q and complex covariance G .

$$\text{i.e. } G_{ij} = \langle [\hat{C}_i^\dagger, \hat{C}_j] \rangle = 2\langle \hat{C}_i^\dagger \hat{C}_j \rangle - \delta_{ij}$$

$$= 2C_{ij} - \delta_{ij}, \text{ where } C_{ij} = \langle \hat{C}_i^\dagger \hat{C}_j \rangle \leftarrow \text{Two-point function}$$

Properties: $G^2 = 1$, $\text{Tr}(G) = 2q - N$

Let $\hat{C}_{i=1,\dots,N}^\dagger$ denote local fermion modes.

Define $|G\rangle = \prod_{i=1}^q \hat{X}_i^\dagger |vac\rangle = \hat{X}_1^\dagger \hat{X}_2^\dagger \dots \hat{X}_q^\dagger |vac\rangle$, where $q=0,\dots,N$ is the total charge

Here, $\hat{X}_{i=1,\dots,N}^\dagger$ are a collection of quasiparticle operators defined as

$$\hat{X}_i^\dagger = \sum_{j=1}^N \hat{C}_j^\dagger U_{ji}, \quad \hat{X}_i = \sum_{j=1}^N (U^\dagger)_{ij} \hat{C}_j, \text{ where } U \in U(N).$$

Accordingly: $\hat{C}_i^\dagger = \sum_{j=1}^N \hat{X}_j^\dagger (U^\dagger)_{ji}$, $\hat{C}_i = \sum_{j=1}^N U_{ij} \hat{X}_j$

How does an excitation evolve? i.e. $\hat{C}_j^\dagger |G\rangle \leftarrow$ How do we write this in terms of G ?

First, consider $\langle G | \hat{C}_i^\dagger \hat{C}_j | G \rangle = \sum_{i',j'} \langle G | \hat{X}_{i'}^\dagger (U^\dagger)_{i'i} (U)_{jj'} \hat{X}_{j'} | G \rangle$

$$= \sum_{i',j'} (U^\dagger)_{i'i} (U)_{jj'} \underbrace{\langle G | \hat{X}_{i'}^\dagger \hat{X}_{j'} | G \rangle}$$

$$= D_{i'j'} \propto \delta_{i'j'}, \quad D = \begin{pmatrix} 1_q & \\ & 0_{N-q} \end{pmatrix}$$

$$\Rightarrow \underline{C_{ij} = (U^* D U^T)_{ij}}, \text{ then } G = 2C - 1$$

$$\text{Now, consider } \hat{X}_i^\dagger |G\rangle = \begin{cases} 0, & \text{if } i \in \{1, \dots, q\} \\ |G'\rangle \neq 0, & \text{else} \end{cases} \Rightarrow \hat{X}_i^\dagger |G\rangle = \sum_j (1-D)_{ij} \hat{X}_j^\dagger |G\rangle$$

$$\text{It follows, } \sum_j \hat{C}_j^\dagger U_{ji} |G\rangle = \sum_{j'} (1-D)_{ij} \hat{C}_{j'}^\dagger U_{j'j} |G\rangle$$

$$\Rightarrow U^T \hat{C}^\dagger |G\rangle = (1-D) U^T \hat{C}^\dagger |G\rangle$$

$$\Rightarrow \hat{C}^\dagger |G\rangle = (1-C) \hat{C}^\dagger |G\rangle = (1 - \frac{1}{2}(G+1)) \hat{C}^\dagger |G\rangle = \frac{1}{2}(1-G) \hat{C}^\dagger |G\rangle, \quad C_{ij} = (U^* D U^T)_{ij} \checkmark$$

$$\Rightarrow \underline{\hat{C}_i^\dagger |G\rangle = - \sum_j G_{ij} \hat{C}_j^\dagger |G\rangle}$$

Similarly, We can track a hole excitation

$$\hat{X}_i |G\rangle = \begin{cases} |G''\rangle \neq 0, & \text{else} \\ 0, & \text{if } i \in \{N-2, \dots, N\} \end{cases} \Rightarrow \hat{X}_i |G\rangle = \sum_j D_{ij} \hat{X}_j |G\rangle$$

$$\text{Then, } \sum_{j=1}^N (u^\dagger)_{ij} \hat{C}_j |G\rangle = \sum_{jj'} D_{ij} (u^\dagger)_{jj'} \hat{C}_{j'} |G\rangle$$

$$u^\dagger \hat{C} |G\rangle = D u^\dagger \hat{C} |G\rangle \Rightarrow \hat{C} |G\rangle = u D u^\dagger \hat{C} |G\rangle = C^T \hat{C} |G\rangle$$

$$\Rightarrow \hat{C}_i |G\rangle = \sum_j C_{ij} \hat{C}_j |G\rangle = \sum_j \frac{1}{2} (G_{ij}^T + \delta_{ij}) \hat{C}_j |G\rangle$$

$$\Rightarrow \underline{\hat{C}_i |G\rangle = \sum_j G_{ij}^T \hat{C}_j |G\rangle}$$

In summary

$$\hat{C}_i^\dagger |G\rangle = - \sum_j G_{ij} \hat{C}_j^\dagger |G\rangle$$

$$\hat{C}_i |G\rangle = \sum_j G_{ij}^T \hat{C}_j |G\rangle$$

$$G^T = G^*$$

$$\text{Bra relations } \langle G | \hat{C}_i = - \langle G | \sum_j G_{ij}^* \hat{C}_j = - \sum_j \langle G | \hat{C}_j G_{ji}$$

$$\langle G | \hat{C}_i^\dagger = \sum_j \langle G | \hat{C}_j^\dagger G_{ij} = \sum_j \langle G | \hat{C}_j^\dagger G_{ji}^*$$

$$\text{Check: } \langle G | \hat{C}_i^\dagger \hat{C}_j |G\rangle = \sum_{i'j'} \langle G | \hat{C}_{i'}^\dagger G_{i'i}^* G_{j'j}^* \hat{C}_{j'} |G\rangle$$

$$= \sum_{i'j'} G_{i'i}^* G_{j'j}^* \langle G | \hat{C}_{i'}^\dagger \hat{C}_{j'} |G\rangle$$

$$= \sum_{i'j'} G_{i'i}^* G_{j'j}^* G_{i'j'} = \sum_{i'j'} G_{ii'} G_{jj'} G_{i'j'} = G_{ij} \checkmark$$

Mixed State Formalism (State - covariance Relation)

Consider a Gaussian mixed state characterized by complex covariance G

$$\dot{\epsilon}_G = \frac{1}{2} e^{-\lambda \Sigma_{ij} h_{ij} \dot{c}_i^\dagger \dot{c}_j}, \quad G = -\text{Tanh}(\lambda h^T/2), \quad \lambda \in \mathbb{R}, |\text{spec}(G)| < 1$$

Suppose that $\dot{\epsilon}_G$ is full-rank, such that $\dot{\epsilon}_G^{-1}$ is well-defined.

Consider the left/right action of a creation operator (use Einstein summation)

$$\text{left: } \dot{c}_k^\dagger \dot{\epsilon}_G = \dot{\epsilon}_G \dot{\epsilon}_G^{-1} \dot{c}_k^\dagger \dot{\epsilon}_G = \dot{\epsilon}_G (e^{\lambda h^T})_{kl} \dot{c}_l^\dagger \quad (*)$$

$$\text{right: } \dot{\epsilon}_G \dot{c}_k^\dagger = \dot{\epsilon}_G \dot{c}_k^\dagger \dot{\epsilon}_G^{-1} \dot{\epsilon}_G = (e^{-\lambda h^T})_{kl} \dot{c}_l^\dagger \dot{\epsilon}_G \quad (**)$$

$$(*) + (**): \dot{c}_k^\dagger \dot{\epsilon}_G + \dot{\epsilon}_G \dot{c}_k^\dagger = \dot{\epsilon}_G (e^{\lambda h^T})_{kl} \dot{c}_l^\dagger + (e^{-\lambda h^T})_{kl} \dot{c}_l^\dagger \dot{\epsilon}_G$$

$$\Rightarrow [\dot{c}_k^\dagger - (e^{-\lambda h^T})_{kl}] \dot{\epsilon}_G = -\dot{\epsilon}_G [\dot{c}_k^\dagger - (e^{\lambda h^T})_{kl} \dot{c}_l^\dagger]$$

$$\text{Now use } e^{\pm \lambda h^T} = e^{\mp 2 \arctanh(G)} = \frac{1 \mp G}{1 \pm G}$$

$$\Rightarrow [\dot{c}_k^\dagger - \left(\frac{1+G}{1-G}\right)_{kl} \dot{c}_l^\dagger] \dot{\epsilon}_G = -\dot{\epsilon}_G [\dot{c}_k^\dagger - \left(\frac{1-G}{1+G}\right)_{kl} \dot{c}_l^\dagger]$$

multiply both sides by $(1-G^2)$

$$\Rightarrow [(1-G^2)_{kl} \dot{c}_l^\dagger - (1+G)_{kl}^2 \dot{c}_l^\dagger] \dot{\epsilon}_G = -\dot{\epsilon}_G [(1-G^2)_{kl} \dot{c}_l^\dagger - (1-G)_{kl}^2 \dot{c}_l^\dagger]$$

$$1-G^2 - 1-2G-G^2 = -2G[G+1], \quad 1-G^2 - 1+2G-G^2 = -2G[G-1]$$

$$\Rightarrow [(G(G+1))_{kl} \dot{c}_l^\dagger] \dot{\epsilon}_G = -\dot{\epsilon}_G [G(G-1)_{kl} \dot{c}_l^\dagger]$$

Does this agree with pure state limit?

$$\dot{\epsilon}_G \rightarrow |G\rangle \langle G|, \quad G^2=1 \Rightarrow G(G+1) = 1+G, \quad G(G-1) = 1-G$$

$$\Rightarrow (G+1)_{kl} \dot{c}_l^\dagger |G\rangle \langle G| = |G\rangle \langle G| (G-1)_{kl} \dot{c}_l^\dagger, \quad \cdot |G\rangle \text{ on both sides}$$

$$\Rightarrow \dot{c}_k^\dagger |G\rangle = -G_{kl} \dot{c}_l^\dagger |G\rangle \quad \checkmark, \quad \text{apply } \langle G| \cdot \text{ on both sides}$$

$$\langle G| \dot{c}_k^\dagger = \langle G| G_{kl} \dot{c}_l^\dagger \quad \checkmark$$

$$\text{Thus, } \dot{\epsilon}_G \text{ satisfies } \underline{[(G(G+1))_{kl} \dot{c}_l^\dagger] \dot{\epsilon}_G = -\dot{\epsilon}_G [G(G-1)_{kl} \dot{c}_l^\dagger]}$$

What about the analogous hole relation?

$$\text{left: } \dot{c}_k \dot{e}_G = \dot{e}_G \dot{e}_G^{-1} \dot{c}_k \dot{e}_G = \dot{e}_G (e^{-\lambda h})_{kl} \dot{c}_l^\dagger \quad (*)$$

$$\text{right: } \dot{e}_G \dot{c}_k = \dot{e}_G \dot{c}_k \dot{e}_G^{-1} \dot{e}_G = (e^{\lambda h})_{kl} \dot{c}_l \dot{e}_G \quad (**)$$

$$(*) + (**): \dot{c}_k \dot{e}_G + \dot{e}_G \dot{c}_k = \dot{e}_G (e^{-\lambda h})_{kl} \dot{c}_l + (e^{\lambda h})_{kl} \dot{c}_l^\dagger \dot{e}_G$$

$$\Rightarrow [\dot{c}_k - (e^{\lambda h})_{kl} \dot{c}_l] \dot{e}_G = -\dot{e}_G [\dot{c}_k - (e^{-\lambda h})_{kl} \dot{c}_l]$$

$$\text{use } e^{\pm \lambda h} = \frac{1 \mp G^T}{1 \pm G^T}$$

$$\Rightarrow [\dot{c}_k - \left(\frac{1-G^T}{1+G^T}\right)_{kl} \dot{c}_l] \dot{e}_G = -\dot{e}_G [\dot{c}_k - \left(\frac{1+G^T}{1-G^T}\right)_{kl} \dot{c}_l]$$

$$\Rightarrow [(1-G^{T^2})_{kl} \dot{c}_l - (1-G^T)^2_{kl} \dot{c}_l] \dot{e}_G = -\dot{e}_G [(1-G^{T^2})_{kl} \dot{c}_l - (1+G^T)^2_{kl} \dot{c}_l]$$

$$\Rightarrow [-2(G^{T^2} - G^T)_{kl} \dot{c}_l] \dot{e}_G = -\dot{e}_G [-2(G^{T^2} + G^T)_{kl} \dot{c}_l]$$

$$\Rightarrow \underline{[G^T(G^T - 1)]_{kl} \dot{c}_l \dot{e}_G = -\dot{e}_G [G^T(G^T + 1)]_{kl} \dot{c}_l}$$

Pure-state limit?

$$\text{left: } (1 - G^T)_{kl} \dot{c}_l |G\rangle = 0 \Rightarrow \dot{c}_k |G\rangle = G^T_{kl} \dot{c}_l |G\rangle \quad \checkmark$$

$$\text{right: } \langle G | (1 + G^T)_{kl} \dot{c}_l = 0 \Rightarrow \langle G | \dot{c}_k = -G^T_{kl} \dot{c}_l \langle G | \quad \checkmark$$

Thus, the mixed state relations are

$$[G(G+1)]_{kl} \dot{c}_l^\dagger \dot{e}_G = -\dot{e}_G [G(G-1)]_{kl} \dot{c}_l^\dagger$$

$$[G^T(G^T - 1)]_{kl} \dot{c}_l \dot{e}_G = -\dot{e}_G [G^T(G^T + 1)]_{kl} \dot{c}_l$$

Transfer Matrix Formalism (pure state)

1) Full-Rank Gaussian Operator

The most generic $U(1)$ -Conserving full-rank Gaussian operator can be written as

$$\hat{V} = e^{\sum_{ij} \lambda M_{ij} \hat{c}_i^\dagger \hat{c}_j}, \text{ where } M \in \mathbb{C}^{N \times N} \text{ is full-rank and } |\text{spec}(MM^\dagger)| < \infty$$

The particle transfer matrix T_p is defined via

$$\begin{aligned} \hat{V} \hat{c}_\kappa^\dagger \hat{V}^{-1} &= \hat{c}_\kappa^\dagger + \lambda \sum_{ij} M_{ij} [\hat{c}_i^\dagger \hat{c}_j, \hat{c}_\kappa^\dagger] + O(\lambda^2), \quad [\hat{c}_i^\dagger \hat{c}_j, \hat{c}_\kappa^\dagger] = \delta_{j\kappa} \hat{c}_i^\dagger \\ &= \hat{c}_\kappa^\dagger + \lambda \sum_i M_{\kappa i}^\top \hat{c}_i^\dagger + O(\lambda^2) \end{aligned}$$

$$\Rightarrow \hat{V} \hat{c}_\kappa^\dagger \hat{V}^{-1} = \sum_j (e^{\lambda M^\top})_{\kappa j} \hat{c}_j^\dagger \Rightarrow \underline{T_p = e^{\lambda M^\top} \in GL(\mathbb{C}, N)}$$

Similarly, the hole transfer matrix T_h is defined via

$$\begin{aligned} \hat{V} \hat{c}_\kappa \hat{V}^{-1} &= \hat{c}_\kappa + \lambda \sum_{ij} M_{ij} [\hat{c}_i^\dagger \hat{c}_j, \hat{c}_\kappa] + O(\lambda^2), \quad [\hat{c}_i^\dagger \hat{c}_j, \hat{c}_\kappa] = -\delta_{i\kappa} \hat{c}_j \\ &= \hat{c}_\kappa - \lambda \sum_j M_{\kappa j} \hat{c}_j + O(\lambda^2) \end{aligned}$$

$$\Rightarrow \hat{V} \hat{c}_\kappa \hat{V}^{-1} = \sum_j (e^{-\lambda M})_{\kappa j} \hat{c}_j \Rightarrow \underline{T_h = e^{-\lambda M} \in GL(\mathbb{C}, N)}$$

Therefore, the particle and hole transfer matrices are related via

$$\underline{T_p = (T_h)^{-T}}$$

Aside: polar decompositions. We can always decompose $\hat{V}$ in terms of a Hermitian PSD Gaussian operator and a Gaussian unitary

i.e. $\hat{V} = \hat{U}_u \hat{K}$, where $\hat{K} = \hat{K}^\dagger = e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$, with $H = H^\dagger$ and $|\text{spec}(H)| < \infty$,

and where $\hat{U}_u \hat{c}_\kappa^\dagger \hat{U}_u^{-1} = \sum_j (u^\top)_{\kappa j} \hat{c}_j^\dagger$; $\hat{U}_u \hat{c}_\kappa \hat{U}_u^{-1} = \sum_j (u^\dagger)_{\kappa j} \hat{c}_j$, where $u \in U(N)$

Then, the transfer matrix can be decomposed as

$$\underline{T_p = e^{\lambda H^\top} u^\top, \quad T_h = e^{-\lambda H} u^\dagger \quad (\text{with } T_p = (T_h)^{-T})$$

2) Choi - Covariance Representation

Every operator has a state representation via Choi - Isomorphism.

If the operator and reference maximally-entangled state are Gaussian, then the Choi-representation of the operator is completely characterized by a covariance matrix.

Define the reference state over $\mathcal{H}_L \otimes \mathcal{H}_R$ as

$$|REF\rangle_{LR} = \frac{1}{\sqrt{2}} [|0\rangle_L |1\rangle_R + |1\rangle_L |0\rangle_R]$$

Such that it has strong global $U(1)$ -symmetry.

The operator $\hat{V}: \mathcal{H} \rightarrow \mathcal{H}$ is vectorized on a doubled Hilbert space

$\hat{V} \rightarrow |\Sigma\rangle_{LR} \equiv \hat{V}_R |REF\rangle_{LR} \in \mathcal{H}_L \otimes \mathcal{H}_R$, where we fix the convention

$\mathcal{H}_L = \mathcal{H}$, $\mathcal{H}_R = \bar{\mathcal{H}}$ (i.e. for $|\psi\rangle \in \mathcal{H} \Rightarrow |\bar{\psi}\rangle \in \mathcal{H}$, where $\bar{(\cdot)}$ is binary complement)

i.e. physical states are mapped from $\mathcal{H}_L \rightarrow \mathcal{H}_R$

Furthermore, Σ is the complex covariance representation of $\hat{V}_R |REF\rangle_{LR}$

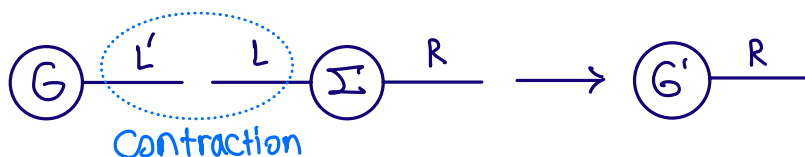
$$\text{i.e. } \Sigma_{ij, \alpha\beta} = \langle \Sigma | [\hat{C}_{i,\alpha}^\dagger, \hat{C}_{j,\beta}] | \Sigma \rangle_{LR}, \quad ij=1, \dots, N, \quad \alpha\beta \in \{L, R\}$$

We can parametrize Σ in terms of blocks on LR: $\Sigma = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{RL} & \Sigma_{RR} \end{pmatrix}$

Hermiticity: $\Sigma = \Sigma^\dagger \Rightarrow \Sigma_{\alpha\alpha} = \Sigma_{\alpha\alpha}^\dagger, \Sigma_{\alpha\beta} = \Sigma_{\beta\alpha}^\dagger, \alpha\beta \in \{L, R\}$

Pure State condition: $\Sigma^2 = 1 \Rightarrow \begin{cases} \Sigma_{LL}^2 + \Sigma_{LR}\Sigma_{RL} = 1 \\ \Sigma_{RR}^2 + \Sigma_{RL}\Sigma_{LR} = 1 \\ \Sigma_{LL}\Sigma_{LR} + \Sigma_{LR}\Sigma_{RR} = 0 \end{cases}$

Evolution Convention: $V_\Sigma |G\rangle \propto |G'\rangle \Rightarrow \langle REF | (|G\rangle_L | \Sigma \rangle_{LR}) \propto |G'\rangle_R$



3) Transfer-Matrix - Choi Covariance mapping

Suppose we have a Choi-state representation of an operator $\hat{V}$

$$\text{i.e. } |\Sigma\rangle_{LR} = \hat{V}_R |\text{REF}\rangle_{LR}$$

How is Σ related to particle and hole-sector transfer matrices?

$$\text{Useful fact: } \hat{C}_{i,L} |\text{REF}\rangle_{LR} = \hat{C}_{i,R} |\text{REF}\rangle_{LR}, \quad \hat{C}_{i,L}^\dagger |\text{REF}\rangle_{LR} = -\hat{C}_{i,R}^\dagger |\text{REF}\rangle_{LR}$$

$$\text{Then, } \hat{C}_{i,L}^\dagger |\Sigma\rangle_{LR} = \hat{C}_{i,L}^\dagger \hat{V}_R |\text{REF}\rangle_{LR} = -\hat{V}_R \hat{C}_{i,R}^\dagger |\text{REF}\rangle_{LR} = -\sum_j (T_p)_{ij} \hat{C}_{j,R}^\dagger |\Sigma\rangle_{LR}$$

$$\hat{C}_{i,L} |\Sigma\rangle_{LR} = \hat{C}_{i,L} \hat{V}_R |\text{REF}\rangle_{LR} = \hat{V}_R \hat{C}_{i,R} |\text{REF}\rangle_{LR} = \sum_j (T_h)_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR}$$

To understand the map between $T_{p,h}$ to Σ , we apply the state relations

$$\text{particle: } \hat{C}_{i,L}^\dagger |\Sigma\rangle_{LR} = -(\Sigma_{LL})_{ij} \hat{C}_{j,L}^\dagger |\Sigma\rangle_{LR} - (\Sigma_{LR})_{ij} \hat{C}_{j,R}^\dagger |\Sigma\rangle_{LR} \quad (\text{Einstein summation})$$

$$\Rightarrow \hat{C}_{i,L}^\dagger |\Sigma\rangle_{LR} = -[(1 + \Sigma_{LL})^{-1} \Sigma_{LR}]_{ij} \hat{C}_{j,R}^\dagger |\Sigma\rangle_{LR} = -(T_p)_{ij} \hat{C}_{j,R}^\dagger |\Sigma\rangle_{LR}$$

$$\text{Thus, } \underline{T_p = (1 + \Sigma_{LL})^{-1} \Sigma_{LR}},$$

Using the purity condition

$$\Sigma_{LL} \Sigma_{LR} = -\Sigma_{LR} \Sigma_{RR} \Rightarrow \Sigma_{LR}^{-1} \Sigma_{LL} = -\Sigma_{RR} \Sigma_{LR}^{-1}, \text{ we equivalently have}$$

$$T_p = (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = (\Sigma_{LR}^{-1} + \Sigma_{LR}^{-1} \Sigma_{LL})^{-1} = \Sigma_{LR} (1 - \Sigma_{RR})^{-1}$$

$$\text{hole: } \hat{C}_{i,L} |\Sigma\rangle_{LR} = (\Sigma_{LL}^T)_{ij} \hat{C}_{j,L} |\Sigma\rangle_{LR} + (\Sigma_{RL}^T)_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR}$$

$$\Rightarrow \hat{C}_{i,L} |\Sigma\rangle_{LR} = [(1 - \Sigma_{LL}^T) \Sigma_{RL}^T]_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR} = (T_h)_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR}$$

$$\text{Thus, } \underline{T_h = (1 - \Sigma_{LL}^T)^{-1} \Sigma_{RL}^T} = \Sigma_{RL}^T (1 + \Sigma_{RR}^T) \quad (\text{Via purity condition})$$

Since $T_p = T_h^{-T}$, we must verify that

$$(1 + \Sigma_{LL})^{-1} \Sigma_{LR} = [(1 - \Sigma_{LL}^T)^{-1} \Sigma_{RL}^T]^{-T} = (1 - \Sigma_{LL}) \Sigma_{RL}^{-1}$$

$$\text{This is guaranteed by purity: } (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = (1 - \Sigma_{LL}) \Sigma_{RL}^{-1}$$

$$\Rightarrow \Sigma_{LR} \Sigma_{RL} = 1 - \Sigma_{LL}^2 \quad \checkmark \quad (\text{True via purity})$$

In summary,
$$\begin{cases} T_p = (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = \Sigma_{LR} (1 - \Sigma_{RR})^{-1} \\ T_h = (1 - \Sigma_{LL}^T)^{-1} \Sigma_{RL}^T = \Sigma_{RL}^T (1 + \Sigma_{RR}^T) \end{cases}$$

What is the inverse map? Start with particle-sector

Well, $\Sigma_{LR} = (1 + \Sigma_{LL}) T_p$, $\Sigma_{LR} \Sigma_{LR}^\dagger = 1 - \Sigma_{LL}^2$

$$\Rightarrow (1 + \Sigma_{LL}) T_p T_p^\dagger (1 + \Sigma_{LL}) = 1 - \Sigma_{LL}^2 = (1 - \Sigma_{LL})(1 + \Sigma_{LL})$$

$$\Rightarrow T_p T_p^\dagger + \Sigma_{LL} T_p T_p^\dagger = 1 - \Sigma_{LL} \Rightarrow \Sigma_{LL} = \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger}$$

Then $\Sigma_{LR} = \left[1 + \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} \right] T_p \Rightarrow \Sigma_{LR} = \frac{2}{1 + T_p T_p^\dagger} T_p$

Furthermore, $\Sigma_{LL} \Sigma_{LR} + \Sigma_{LR} \Sigma_{RR} = 0 \Rightarrow \Sigma_{RR} = -\Sigma_{LR}^{-1} \Sigma_{LL} \Sigma_{LR}$

$$\Rightarrow \Sigma_{RR} = -\cancel{\frac{1}{2} T_p^{-1} (1 + T_p T_p^\dagger)} (1 + T_p T_p^\dagger)^{-1} (1 - T_p T_p^\dagger) (1 + T_p T_p^\dagger)^{-1} \cancel{2} T_p$$

$$\Rightarrow \Sigma_{RR} = -T_p^{-1} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} T_p$$

Thus, the full Choi-covariance can be written as

$$\Sigma = \begin{pmatrix} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} & \frac{2}{1 + T_p T_p^\dagger} T_p \\ T_p^\dagger \frac{2}{1 + T_p T_p^\dagger} & -T_p^{-1} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} T_p \end{pmatrix} \quad T_p^{-1} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} T_p = \frac{1 - T_p^\dagger T_p}{1 + T_p^\dagger T_p}$$

Now, decompose $T_p = e^{\lambda H^T} U^T \Leftrightarrow \hat{V} = \hat{U}_u e^{\lambda \Sigma_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$

Then, $\frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} = \frac{1 - e^{2\lambda H^T}}{1 + e^{2\lambda H^T}} = \frac{e^{-\lambda H^T} - e^{\lambda H^T}}{e^{-\lambda H^T} + e^{\lambda H^T}} = -\tanh(\lambda H^T)$

$$T_p^{-1} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} T_p = -U^* \tanh(\lambda H^T) U^T, \quad \frac{2}{1 + T_p T_p^\dagger} T_p = \text{sech}(\lambda H^T) U^T$$

Thus, we find the following set of relations for full-rank $\hat{V}$

$$\hat{V} = \hat{U}_u e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j} \longleftrightarrow T_\rho = e^{\lambda H^T} U^T = T_h^{-T}$$

$$\Sigma = \begin{pmatrix} -\text{Tanh}(\lambda H^T) & \text{sech}(\lambda H^T) U^T \\ U^* \text{sech}(\lambda H^T) & U^* \text{Tanh}(\lambda H^T) U^T \end{pmatrix}$$

4) Choi-Covariance Description of Rank-1 Kraus operators

Consider $\hat{K}_\sigma = e^{\sigma \lambda \hat{N}}$, where $\hat{N} = \hat{\chi}^\dagger \hat{\chi}$, $\hat{\chi}^\dagger = \sum_i \chi_i^* \hat{c}_i^\dagger$, $\sigma = \pm$ different from typical convention

$\hat{N} = \hat{\chi}^\dagger \hat{\chi} = \sum_{ij} \chi_i^* \chi_j \hat{c}_i^\dagger \hat{c}_j = \sum_{ij} (P^T)_{ij} \hat{c}_i^\dagger \hat{c}_j$, where $P = \chi \chi^\dagger$ is single-particle rank-1.

The associated (particle-sector) transfer matrix is

$$T_\rho = e^{\sigma \lambda P} = (1-P) + e^{\sigma \lambda} P, \quad T_\rho = T_\rho^\dagger$$

While the associated Choi-covariance is

$$\Sigma = \begin{pmatrix} -\text{Tanh}(\sigma \lambda P) & \text{sech}(\lambda P) \\ \text{sech}(\lambda P) & \text{Tanh}(\sigma \lambda P) \end{pmatrix}$$

We now take the projective limit: $\lambda \rightarrow \infty$

At the many-body level,

$$\lim_{\lambda \rightarrow \infty} \frac{e^{\sigma \lambda \hat{N}}}{Z_\sigma(\lambda)} \longrightarrow \begin{cases} \hat{N}, & \sigma = + \\ 1 - \hat{N}, & \sigma = - \end{cases}$$

At the transfer-matrix level,

$$\lim_{\lambda \rightarrow \infty} e^{\sigma \lambda P} \longrightarrow \begin{cases} \lim_{\lambda \rightarrow \infty} e^{\lambda} P + (1-P), & \sigma = + \\ 1-P, & \sigma = - \end{cases}$$

At the Choi-covariance level,

$$\Sigma = \begin{pmatrix} -\sigma \tanh(\lambda P) & \text{sech}(\lambda P) \\ \text{sech}(\lambda P) & \sigma \tanh(\lambda P) \end{pmatrix} \xrightarrow{\lambda \rightarrow \infty} \begin{pmatrix} -\sigma P & 1-P \\ 1-P & \sigma P \end{pmatrix}, \sigma = \pm$$

Notice that even if the transfer matrix is divergent, the Choi-covariance remains well-defined in the projective limit.

Let us check this:

$$T_p = (1 + \Sigma_{LL})^{-1} \Sigma_{LR} = (1 - \sigma \tanh(\lambda P))^{-1} \text{sech}(\lambda P) \sim \begin{pmatrix} 1 - \sigma \tanh(\lambda) & \\ & 1 \end{pmatrix}^{-1} \begin{pmatrix} \text{sech}(\lambda) \\ \\ \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\text{sech}(\lambda)}{1 - \sigma \tanh(\lambda)} & \\ & 1 \end{pmatrix} \xrightarrow{\lambda \rightarrow \infty} \begin{cases} \begin{pmatrix} \infty & \\ & 1 \end{pmatrix}, \sigma = + \\ \begin{pmatrix} 0 & \\ & 1 \end{pmatrix}, \sigma = - \end{cases} \sim \lim_{\lambda \rightarrow \infty} [e^{\sigma \lambda} P + (1-P)]$$

The transfer matrix blows up as soon as $1 + \Sigma_{LL}$ becomes rank-deficient.

Identically, when $1 - \Sigma_{RR}$ becomes rank-deficient. This is precisely the scenario where Σ_{RR} has at least one fully occupied mode.

Let's check the hole-sector too.

$$T_h = (1 - \Sigma_{LL}^T)^{-1} \Sigma_{LR}^T = (1 + \sigma \tanh(\lambda) P^T) \text{sech}(\lambda P^T)$$

$$\sim \begin{pmatrix} \frac{\text{sech}(\lambda)}{1 + \sigma \tanh(\lambda)} & \\ & 1 \end{pmatrix} \xrightarrow{\lambda \rightarrow \infty} \begin{cases} \begin{pmatrix} 0 & \\ & 1 \end{pmatrix}, \sigma = + \\ \begin{pmatrix} \infty & \\ & 1 \end{pmatrix}, \sigma = - \end{cases} \sim \lim_{\lambda \rightarrow \infty} [e^{-\sigma \lambda} P^T + (1-P^T)]$$

$$\Rightarrow \underline{T_h = T_p^{-T}}$$

5) Transfer Matrices and fermion counting

So far we have considered dynamics associated with fermion-bilinears. What about single-fermion Kraus operators?

5a) Gain Processes

Let $\hat{V}$ denote a Gaussian operator built from fermion bilinears.

Consider the gain process: $\hat{X}^\dagger \hat{V}$. Since $\hat{X}^\dagger = \hat{X}^\dagger \hat{X} \hat{X}^\dagger = \hat{X}^\dagger (1 - \hat{N})$, the operator $\hat{V}$ has gauge ambiguity i.e.

For $\hat{X}^\dagger \hat{V}$, $\hat{V} \sim \hat{V}' = (1 - \hat{N}) \hat{V} + \hat{N} \hat{\Phi}$, where $\hat{\Phi}$ is any generic operator.

$$\Rightarrow \hat{X}^\dagger \hat{V} = \hat{X}^\dagger \hat{V}'$$

Let's fix the gauge such that $\hat{V} = (1 - \hat{N}) \hat{V}$. It's associated transfer matrix is then ill-defined, but we'll deal with that later.

The solⁿ is: $\hat{V} = (1 - \hat{N}) \hat{K}$, $\hat{K} = e^{\lambda \sum_{ij} M_{ij} \hat{c}_i^\dagger \hat{c}_j} = \prod_u e^{\lambda \sum_{ij} H_{ij} \hat{c}_i^\dagger \hat{c}_j}$

Define $|\mathcal{I}'\rangle_{LR} = \hat{X}_R^\dagger |\mathcal{I}\rangle_{LR}$, $|\mathcal{I}\rangle_{LR} = \hat{V}_R |\text{REF}\rangle_{LR}$

Gauge-Fixed Condition: $|\mathcal{I}\rangle_{LR} = (1 - \hat{N}_R) |\mathcal{I}\rangle_{LR} = \hat{X}_R \hat{X}_R^\dagger |\mathcal{I}\rangle_{LR}$

To regulate the transfer matrix, we write $\hat{V}$ in the limiting form

$$\hat{V} \equiv \lim_{\mu \rightarrow \infty} \hat{V}(\mu) = \lim_{\mu \rightarrow \infty} e^{-\mu \hat{N}} \hat{K}$$

Let $T'_{p,h}$ denote the gauge-fixed Transfer matrix associated with $\hat{V}$

Let $T_{p,h}$ denote the gauge-dependent Transfer matrix associated with $\hat{K}$

Let $T_{p,h}^{\text{eff}}$ denote the gauge-fixed transfer matrix associated with $\hat{X}^\dagger \hat{V}$

$$\begin{aligned} \text{particle propagation} \quad & \hat{c}_{i,L}^\dagger |\mathcal{I}'\rangle_{LR} = \hat{c}_{i,L}^\dagger \hat{X}_R^\dagger |\mathcal{I}\rangle_{LR} = -\hat{X}_R \hat{c}_{i,L}^\dagger |\mathcal{I}\rangle_{LR} = \hat{X}_R^\dagger (T'_p)_{ij} \hat{c}_{j,R}^\dagger |\mathcal{I}\rangle_{LR} \\ & \Rightarrow \hat{c}_{i,L}^\dagger |\mathcal{I}'\rangle_{LR} = -(T_p^{\text{eff}})_{ij} \hat{c}_{j,R}^\dagger |\mathcal{I}'\rangle_{LR} \end{aligned}$$

Here, we have $T_p^{\text{eff}} = T'_p = T_p \lim_{\mu \rightarrow \infty} e^{-\mu P} = T_p Q$, $Q \equiv 1 - P$

Thus, $T_p^{\text{eff}} = T_p^{\text{eff}} Q$.

We conclude that a particle excitation propagates as

$$\underline{\hat{C}_{i,L}^\dagger \mapsto -(T_p^{\text{eff}})_{ij} \hat{C}_{j,R}^\dagger, \text{ With } T_p^{\text{eff}} = T_p Q}$$

Thus, the particle excitation only propagates through null-sector of P.

Hole Propagation

$$\begin{aligned} \left| \hat{C}_{i,L} |\Sigma'\rangle_{LR} \right. &= -\dot{\chi}_R^\dagger (T_h)_{ij} \hat{C}_{j,R} |\Sigma\rangle_{LR} = (T_h)_{ij} [\hat{C}_{j,R} |\Sigma'\rangle_{LR} - \chi_j^* |\Sigma\rangle_{LR}] \\ &= (T_h)_{ij} [\hat{C}_{j,R} |\Sigma'\rangle_{LR} - \chi_j^* \dot{\chi}_R |\Sigma'\rangle_{LR}] \\ &= (T_h)_{ij} [\hat{C}_{j,R} - \chi_j^* \chi_K \dot{C}_{K,R}] |\Sigma'\rangle_{LR} \\ &= (T_h)_{ij} (\hat{C}_{j,R} - P_{jK}^T \dot{C}_{K,R}) |\Sigma'\rangle_{LR} \\ \Rightarrow \hat{C}_{i,L} |\Sigma'\rangle_{LR} &= (T_h^{\text{eff}})_{ij} \hat{C}_{j,R}, \text{ With } T_h^{\text{eff}} = T_h' Q^T = T_h \lim_{\mu \rightarrow \infty} e^{\mu P^T} Q^T \end{aligned}$$

Importantly, the Q^T factor obtained from $\dot{\chi}_R$ is NOT regulated, so we conclude

$$\lim_{\mu \rightarrow \infty} e^{\mu P^T} Q^T = Q^T + \lim_{\mu \rightarrow \infty} e^{\mu P^T} \underbrace{P^T Q^T}_{=0} = Q^T$$

Thus, the hole excitation propagates as

$$\underline{\hat{C}_{i,L} \mapsto (T_h^{\text{eff}})_{ij} \hat{C}_{j,R}, \text{ With } T_h^{\text{eff}} = T_h Q^T}$$

Just like the particle excitation, the hole excitation only propagates through the null-sector of P^T . The implication is that we cannot determine the Choi Covariance output on the sector $||$ to P using Transfer matrix data alone.

Also note that the PHS $T_p = T_h^{-T}$ is broken for T_p^{eff} and T_h^{eff} .

Let us reconstruct the Choi Covariance blocks using the data we have.

The particle Transfer matrix is $T_p^{\text{eff}} = T_p Q$. Since $1 + T_p^{\text{eff}} T_p^{\text{eff}^\dagger}$ remains invertible, we can determine Σ'_{LL} and Σ'_{LR} using the usual trick.

$$\underline{\Sigma'_{LL} = \frac{1 - T_p^{\text{eff}} T_p^{\text{eff}^\dagger}}{1 + T_p^{\text{eff}} T_p^{\text{eff}^\dagger}} = \frac{1 - T_p Q T_p^\dagger}{1 + T_p Q T_p^\dagger}}$$

$$\underline{\Sigma'_{LR} = \frac{2}{1 + T_p^{\text{eff}} T_p^{\text{eff}^\dagger}} T_p^{\text{eff}} = \frac{2}{1 + T_p Q T_p^\dagger} T_p Q \Rightarrow \underline{\Sigma'_{LR} Q = \Sigma'_{LR}}}$$

To determine Σ'_{RR} , We first use the purity constraint

$$\Sigma'_{LR} \Sigma'_{RR} + \Sigma'_{LL} \Sigma'_{LR} = 0 \quad (*)$$

Decompose $\Sigma'_{LR} = \tilde{\Sigma}'_{LR} Q$, Where $\tilde{\Sigma}'_{LR} = \frac{2}{1 + T_p Q T_p^\dagger} T_p$,

Such that $\tilde{\Sigma}'_{LR}$ remains well-defined.

Then (*) becomes: $\tilde{\Sigma}'_{LR} Q \Sigma'_{RR} = -\Sigma'_{LL} \tilde{\Sigma}'_{LR} Q$

$$\Rightarrow Q \Sigma'_{RR} = -\tilde{\Sigma}'_{LR}{}^{-1} \Sigma'_{LL} \tilde{\Sigma}'_{LR} Q \quad (**)$$

Thus, $Q \Sigma'_{RR} Q = -Q \tilde{\Sigma}'_{LR}{}^{-1} \Sigma'_{LL} \tilde{\Sigma}'_{LR} Q$ (left and right multiplication by Q)

Furthermore, $Q \Sigma'_{RR} P = 0$, $\Sigma'_{RR} = \Sigma'_{RR}{}^\dagger \Rightarrow P \Sigma'_{RR} Q = 0$

Thus, we have the decomposition

$$\begin{aligned} \Sigma'_{RR} &= (Q + P) \Sigma'_{RR} (Q + P) = Q \Sigma'_{RR} Q + P \Sigma'_{RR} P \\ &= -Q \tilde{\Sigma}'_{LR}{}^{-1} \Sigma'_{LL} \tilde{\Sigma}'_{LR} Q + P \Sigma'_{RR} P \\ \Rightarrow \Sigma'_{RR} &= -Q T_p^{-1} \frac{1 - T_p Q T_p^\dagger}{1 + T_p Q T_p^\dagger} T_p Q + P \Sigma'_{RR} P \end{aligned}$$

All that's left is to determine $P \Sigma'_{RR} P$.

Using the occupation constraint $\hat{X}_R^\dagger |\Sigma'\rangle_{LR} = 0$,

$$\Rightarrow \hat{X}_R^\dagger \hat{X}_R |\Sigma'\rangle_{LR} = |\Sigma'\rangle_{LR} \Rightarrow \bar{X}^\dagger \Sigma_{RR} \bar{X} = 1 \Rightarrow \underline{P \Sigma_{RR} P = P}$$

Thus, $\Sigma'_{RR} = P - Q T_p^{-1} \frac{1 - T_p Q T_p^\dagger}{1 + T_p Q T_p^\dagger} T_p Q$, and so Σ' is given by:

$$\Sigma' = \begin{pmatrix} \frac{1 - T_p Q T_p^\dagger}{1 + T_p Q T_p^\dagger} & \frac{2}{1 + T_p Q T_p^\dagger} T_p Q \\ Q T_p^\dagger \frac{2}{1 + T_p Q T_p^\dagger} & P - Q \frac{1 - Q T_p^\dagger T_p Q}{1 + Q T_p^\dagger T_p Q} \end{pmatrix}$$

$$Q T_p^{-1} \frac{1 - T_p Q T_p^\dagger}{1 + T_p Q T_p^\dagger} T_p Q$$

$$= Q \frac{1 - Q T_p^\dagger T_p Q}{1 + Q T_p^\dagger T_p Q}$$

$$[Q, Q T_p^\dagger T_p Q] = 0$$

We can rewrite the Choi covariance in terms the gauge-fixed Transfer matrices

$$T_p^{\text{eff}} = T_p Q. \text{ For simplicity, we drop the 'eff'}$$

$$\Rightarrow \Sigma' = \begin{pmatrix} \frac{1 - T_p T_p^\dagger}{1 + T_p T_p^\dagger} & \frac{2}{1 + T_p T_p^\dagger} T_p \\ T_p^\dagger \frac{2}{1 + T_p T_p^\dagger} & P - Q \frac{1 - T_p^\dagger T_p}{1 + T_p^\dagger T_p} Q \end{pmatrix}$$

Where $T_p = T_p Q$ is the transfer matrix associated with $\hat{V} = (1 - \hat{W})\hat{V}$

Notice that, if we had started with $|\Sigma\rangle = \hat{V}_R |\text{REF}\rangle_{LR}$, $\hat{V}_R = (1 - \hat{W}_R)\hat{V}_R$,

We equivalently have

$$\Sigma' = \begin{pmatrix} 1 & \\ & Q \end{pmatrix} \Sigma \begin{pmatrix} 1 & \\ & Q \end{pmatrix} + \begin{pmatrix} 0 & \\ & P \end{pmatrix}$$

Which is exactly equivalent to applying the set-reset map on the 'Right' leg at the many-body level.

$$\mathcal{E}(\hat{e}) = \hat{W}\hat{e}\hat{W} + \hat{X}^\dagger \hat{e} \hat{X} \text{ (set-reset map)}$$

$$\text{i.e. if } \hat{e} \rightarrow \hat{e}_{LR} = \hat{V}_R |\text{REF}\rangle_{LR} \langle \text{REF}| \hat{V}_R^\dagger$$

$$= (1 - \hat{W}_R) \hat{V}_R |\text{REF}\rangle_{LR} \langle \text{REF}| \hat{V}_R^\dagger (1 - \hat{W}_R)$$

$$\text{Then } \mathcal{E}_R(\hat{e}_{LR}) = \underbrace{\hat{W}_R \hat{V}_R |\text{REF}\rangle_{LR} \langle \text{REF}| \hat{V}_R^\dagger \hat{W}_R}_{=0} + \hat{X}_R^\dagger \hat{V}_R |\text{REF}\rangle_{LR} \langle \text{REF}| \hat{V}_R \hat{X}_R$$

$$= \hat{X}_R^\dagger \hat{V}_R |\text{REF}\rangle_{LR} \langle \text{REF}| \hat{V}_R \hat{X}_R \checkmark$$

$$\text{Ex. } T_p = 1, \frac{1-Q}{1+Q} = \frac{P}{2-P} \sim \begin{pmatrix} 1 & \\ & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & \\ & 0 \end{pmatrix} \sim P \Rightarrow Q \frac{1-Q}{1+Q} Q = 0$$

$$Q \frac{2}{1+Q} = (1-P) \frac{2}{2-P} \sim \begin{pmatrix} 0 & \\ & 1 \end{pmatrix} 2 \begin{pmatrix} 1 & \\ & 2 \end{pmatrix}^{-1} \sim Q$$

$$\Rightarrow \Sigma' \Big|_{T_p=1} = \begin{pmatrix} P & Q \\ Q & P \end{pmatrix} \checkmark$$

And the gauge-fixed Transfer matrices are:

$$\begin{cases} T_p^{\text{eff}} = T_p Q, T_p = T_h^{-T}, \text{ but } T_p^{\text{eff}} \neq T_h^{\text{eff}-T} \text{ (PHS broken because of } \chi^\dagger \text{ insertion)} \\ T_h^{\text{eff}} = T_h Q \end{cases}$$

In contrast, without gain feedback, we would have: $\begin{cases} T_p^{\text{eff}} = T_p Q \\ T_h^{\text{eff}} = \lim_{\mu \rightarrow \infty} T_h e^{\mu P^T} \end{cases}$

Importantly, the off-diagonal blocks project out the P-sector and only allow propagation from left to right through the null-sector

$$\hat{C}_{i,L}^\dagger \mapsto -(T_p Q)_{ij} \hat{C}_{j,R}^\dagger$$

The P-sector of the left mode effectively dies. But what about the right mode?

5b) Transfer Matrix Composition under gain

$$\text{Consider now } |\Sigma''\rangle_{LR} = \hat{\chi}_{2,R}^\dagger \hat{K}_{2,R} \hat{\chi}_{1,R}^\dagger \hat{K}_{1,R} |\text{REF}\rangle_{LR}$$

$$\text{Gauge-Redundancy can be handled via } \hat{K}_{2,R} \longrightarrow (1 - \hat{N}_{2,R}) \hat{K}_{2,R} = \lim_{\mu \rightarrow \infty} e^{-\mu \hat{N}_{2,R}} \hat{K}_{2,R}$$

$$\hat{K}_{1,R} \longrightarrow (1 - \hat{N}_{1,R}) \hat{K}_{1,R} = \lim_{\mu \rightarrow \infty} e^{-\mu \hat{N}_{1,R}} \hat{K}_{1,R}$$

$$|\Sigma''\rangle_{LR} = \hat{\chi}_{2,R}^\dagger \hat{K}_{2,R} \hat{\chi}_{1,R}^\dagger \hat{K}_{1,R} |\text{REF}\rangle_{LR}$$

$$= \hat{\chi}_{2,R}^\dagger \hat{\tilde{\chi}}_{1,R}^\dagger \hat{K}_{2,R} \hat{K}_{1,R} |\text{REF}\rangle_{LR}, \quad \tilde{\chi}_k^* = \chi_k^*(T_{p,r})_{kK}$$

$$\hat{\chi}_2^\dagger \hat{\chi}_1^\dagger = \hat{\chi}_2^\dagger (1 - \hat{N}_2) \hat{\chi}_1^\dagger (1 - \hat{N}_1)$$

$$=$$

Choi-Covariance Update Formulae (gain)

Consider a generic Choi-state $|\Sigma\rangle_{LR}$ characterized by a particle transfer matrix $T_p = (1 + \Sigma_L)^{-1} \Sigma_{LR}$ and hole transfer matrix $T_h = T_p^{-T}$.

Now consider a generic quasiparticle $\hat{\chi}^\dagger = \sum_i \chi_i^\dagger \hat{c}_i^\dagger$, $P = \chi \chi^\dagger$

For a quasiparticle gain interaction, how does the Choi-covariance transform?

$$|\Sigma'\rangle_{LR} = \frac{\hat{\chi}_R^\dagger |\Sigma\rangle_{LR}}{\|\hat{\chi}_R^\dagger |\Sigma\rangle_{LR}\|}, \text{ let } C = \frac{1}{2}(\Sigma + 1)$$

$$|\Sigma\rangle = |C\rangle$$

$$\text{Parametrize } \Sigma = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{LR}^\dagger & \Sigma_{RR} \end{pmatrix}$$

$$= \frac{1}{2} - \frac{1}{2} \text{Tr}(\Sigma_{RR} P) = \frac{1}{2} (1 - \text{Tr}(\Sigma_{RR} P))$$

$$\Rightarrow 2p = 1 - \text{Tr}(\Sigma_{RR} P)$$

$$C'_{LL,ij} = \langle \Sigma | \hat{\chi}_R \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{\chi}_R^\dagger | \Sigma \rangle_{LR} / \langle 1 - \hat{N}_R \rangle, \quad \langle 1 - \hat{N}_R \rangle = 1 - \text{Tr}(C_{RR} P) \equiv p$$

$$\langle \Sigma | \hat{\chi}_R \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{\chi}_R^\dagger | \Sigma \rangle_{LR} = \Sigma_{KL} \chi_K \chi_L^* \langle \Sigma | \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{c}_{L,R}^\dagger | \Sigma \rangle_{LR}$$

$$\langle \Sigma | \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{c}_{L,R}^\dagger | \Sigma \rangle_{LR} = \langle \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \rangle \langle \hat{c}_{j,L} \hat{c}_{L,R}^\dagger \rangle + \langle \hat{c}_{K,R} \hat{c}_{L,R}^\dagger \rangle \langle \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \rangle$$

$$= C_{LR,ik} C_{RL,lj} + [\delta_{KL} - C_{RR,lk}] C_{LL,ij}$$

$$\Rightarrow \langle \Sigma | \hat{\chi}_R \hat{c}_{i,L}^\dagger \hat{c}_{j,L} \hat{\chi}_R^\dagger | \Sigma \rangle_{LR} = \Sigma_{KL} \chi_K \chi_L^* [C_{LR,ik} C_{RL,lj} + [\delta_{KL} - C_{RR,lk}] C_{LL,ij}]$$

$$= (C_{LR} P C_{RL})_{ij} + (1 - \text{Tr}(C_{RR} P)) C_{LL,ij}$$

$$\Rightarrow \underline{C'_{LL} = \frac{1}{p} C_{LR} P C_{RL} + C_{LL}}$$

$$C'_{LR,ij} \propto \langle \Sigma | \hat{\chi}_R \hat{c}_{i,L}^\dagger \hat{c}_{j,R} \hat{\chi}_R^\dagger | \Sigma \rangle_{LR} = \Sigma_{KL} \chi_K \chi_L^* \langle \Sigma | \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \hat{c}_{j,R} \hat{c}_{L,R}^\dagger | \Sigma \rangle_{LR}$$

$$= \Sigma_{KL} \chi_K \chi_L^* [\langle \hat{c}_{K,R} \hat{c}_{i,L}^\dagger \rangle \langle \hat{c}_{j,R} \hat{c}_{L,R}^\dagger \rangle + \langle \hat{c}_{K,R} \hat{c}_{L,R}^\dagger \rangle \langle \hat{c}_{i,L}^\dagger \hat{c}_{j,R} \rangle]$$

$$= \Sigma_{KL} \chi_K \chi_L^* [\langle \hat{c}_{i,L}^\dagger \hat{c}_{K,R} \rangle (\langle \hat{c}_{L,R}^\dagger \hat{c}_{j,R} \rangle - \delta_{jL}) + (\delta_{KL} - \langle \hat{c}_{L,R}^\dagger \hat{c}_{K,R} \rangle) \langle \hat{c}_{i,L}^\dagger \hat{c}_{j,R} \rangle]$$

$$= [C_{LR} P (C_{RR} - 1) + (1 - \text{Tr}(C_{RR} P)) C_{LR}]_{ij}$$

$$\Rightarrow \underline{C'_{LR} = \frac{1}{p} C_{LR} P (C_{RR} - 1) + C_{LR}}$$